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## Injuries in Synchronized Skating

### Abstract

Synchronized skating is a relatively new competitive sport and data about injuries in this discipline are lacking. Therefore the purpose of this study was to investigate the frequency and pattern of acute and overuse injuries in synchronized skaters. Before and during the World Synchronized Skating Championship 2004, a questionnaire inquiring about the frequency of injuries in this skating discipline was given to 23 participating teams. A total of 514 women and 14 men senior skaters completed the questionnaires (100% response). Two hundred and eighteen (42.4%) female and 6 (42.9%) male skaters had suffered from acute injuries during their synchronized skating career. As some skaters had suffered from more than one injury, the total number of acute injuries in females was 398 and in males 14. In female skaters 19.8% of acute injuries were head injuries, 7.1% trunk, 33.2% upper, and 39.9% lower extremity injuries. In male skaters 14.3% were head injuries, 28.6% upper, and 57.1% lower extremity injuries, with no report of trunk injuries. Sixty-nine female and 2 male skaters had low back problems and 112 female and 2 male skaters had one or more overuse syndromes during their skating career. Of 155 overuse injuries in female skaters, 102

(65.8%) occurred during their figure skating career, while 53 injuries (34.2%) only occurred when they skated in synchronized skating teams. In male skaters, out of 5 overuse injuries, 4 (80%) occurred in their figure skating career, while 1 (20%) occurred during their synchronized skating career. Out of the total of 412 injuries, 338 (82%) occurred during on-ice practice, while 74 (18%) happened during off-ice training. Ninety-one (26.9%) acute injuries occurred while practicing individual elements, and 247 (73.1%) on-ice injuries occurred while practicing different team elements. We conclude that injuries in synchronized skating should be of medical concern due to an increasing number of acute injuries, especially those that go beyond the soft tissue and include head injuries and fractures. We feel that these more significant injuries *may to some extent be* attributable to the increasing physical demands and technical difficulty required of the teams now participating in a more competitive environment over the last four years.

### Key words

Synchronized skating · acute injuries · overuse syndromes · off-/on-ice training

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### Bibliography

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## Introduction

The International Skating Union (ISU) governs the winter sports division of figure skating, speed skating, short track speed skating, and synchronized skating. Synchronized Skating is the newest discipline within the ISU. In 2000 the first World Synchronized skating Championship was held in Minneapolis, USA with 21 teams from 17 nations, and since then has become an annual event [7].

In synchronized skating up to 20 skaters, mostly females, skate together in unison, performing different formations on the ice. The main characteristic of the sport is synchronized work by skater's legs, body, arms, and head. Synchronization refers to the importance of unison, the accuracy of the formations, and the precision of the team [6].

Junior level skaters are 12 to 19 years old, while Senior skaters are 14 years and over, with no maximum age limit. Junior and Senior teams have Short programs with required elements and Free skating programs while Juveniles, Novices, and Veterans have only the Free skating program performed to music chosen by the team, lasting for a specified period of time. The short program for synchronized skating consists of different required elements: circle, line, block, wheel, intersection, spin, and moves in the field, while the order of elements in the program is optional. The free skating program must be well balanced between all team elements, and the movements in isolation must be linked together harmoniously by a variety of transitions and executed with a minimum of two footed skating. Special credit is given to those teams whose performance includes not only conventional elements performed in clean, fast, and sure synchronization with precise transitions, but also contains innovative and creative elements [6, 7].

To make the sport more attractive and competitive more and more difficult elements are added into the programs. In the last four years spins, jumps, step sequences, movements in isolation, moves in the field and twizzles, turns of one half or entire revolution executed on one or two feet, have been added into synchronized skating programs. Coaches and team managers are trying to maximize the difficulty of these elements in the pursuit of success and podium finishes at leading international competitions and world championships [6].

The rapidly increasing demands on performance of the synchronized skaters may lead to specific training and injury issues. Since epidemiological studies of injuries in elite synchronized skaters are lacking, the aim of this study was to collect data about physical and training characteristics specific to the sport and to assess the occurrence of injuries in senior synchronized skaters. Two types of injuries can be distinguished in all skating disciplines – the acute defined as tissue damage that occurs at a specific time, and overuse injuries, in which the time of occurrence cannot be exactly specified, and in most cases the skaters neither feels, nor remembers the time of initial tissue damage [2, 4, 5]. For this purpose a questionnaire was used to collect data on injuries in this developing ISU skating discipline.

## Materials and Methods

Before the World Synchronized Skating Championship 2004 the questionnaire consisting of 22 questions was mailed to all 23 participating teams, including 528 skaters. The skaters were requested to return the form upon arrival. As not all teams completed the questionnaire before the Event, the questionnaire was redistributed to those teams during Championship held in Zagreb from 1<sup>st</sup> until 4<sup>th</sup> April 2004. After filling out the questionnaire, often with coaches or/and team doctor's assistance, which required 10–25 minutes, all participants were requested to return it before the closure of the Championship to the International Skating Union (ISU) Medical Advisor present at the Championship.

The questionnaires inquired about the site and types of acute and overuse injuries that occurred during their synchronized skating career. It also inquired about the age of the skater, when the injuries occurred, the injury treatment (conservative vs. surgery), and the time period the skaters were required to be off the ice. In addition questions were asked about their current age, the age when they started to skate, as well as the age when they started to skate in synchronized skating. Major parts of the questionnaires inquired about the incidence of injuries with reference to on-/off-ice training, practicing team (i.e. block, line, intersection, etc.) or individual elements (i.e. jumps, spins, etc.) We also inquired about possible overuse syndromes that occurred during their previous figure skating and their synchronized skating career, as well as the age of menarche for female skaters.

After receiving the questionnaires the data were fed into a computer using Excel software. Descriptive statistics was used for analysis of the data.

## Results

The 23 participating teams from 18 countries and 5 continents (Europe, America, Asia, Africa, and Australia) correctly and completely filled in the questionnaires. With a 100% response a total of 514 ladies and 14 men senior skaters completed the questionnaires.

The median age was 19.4 yrs for female skaters (aged between 15 and 28) and 22.2 yrs for male skaters (aged between 18 and 32). Both female and male skaters started to skate when they were between 3 and 14 yrs. Three hundred and ninety-four skaters (75%) who figure skated before joining the synchronized skating team started to skate at an age of 3–7 yrs. Once they joined the team, they did not practice other skating disciplines. One hundred and thirty-two skaters (25%) who initially started their skating career with synchronized skating began to skate between the age of 6 and 14. Surveyed skaters from this research started to train at a competitive level in synchronized skating when they were between 8 and 21 years old. Four hundred and seventy-three skaters (90%) started to compete in synchronized skating (juvenile, novice, junior) when they were between 8 and 15 years old, while only 53 skaters (10%) joined the team in their senior skating career at age between 17 and 21 yrs. The time spent for on-ice practice was variable between the teams. From

**Table 1** Absolute and relative (%) numbers of acute injuries in synchronized skaters at different body parts that were studied

|                                                  | <b>Female<br/>(n = 514)<br/>no. (%)</b> | <b>Male<br/>(n = 14)<br/>no. (%)</b> | <b>Total<br/>(n = 528)<br/>no. (%)</b> |
|--------------------------------------------------|-----------------------------------------|--------------------------------------|----------------------------------------|
| <b>Head</b>                                      | 79 (19.8%)                              | 2 (14.3%)                            | 81 (19.7%)                             |
| <b>Trunk</b>                                     | 28 (7.1%)                               |                                      | 28 (6.8%)                              |
| <b>Upper extremity</b>                           | 132 (33.2%)                             | 4 (28.6%)                            | 136 (33.0%)                            |
| <b>Lower extremity</b>                           | 159 (39.9%)                             | 8 (57.1%)                            | 167 (40.5%)                            |
| <b>Total number of<br/>injuries, no. and (%)</b> | 398 (100%)                              | 14 (100%)                            | 412 (100%)                             |

23 teams, 4 of them spent 10–21 hours (4–6 times) per week participating in on-ice training, while other 20 teams spend 4–8 hours (2–5 times) per week participating in on-ice training. On average, the best placed teams spend 15 hours in on-ice training while other teams spend on average 6 hours in on-ice training. All of the teams spent an additional 2 to 5 hours (3.5 hours in average) per week participating in off-ice training.

Among 514 female synchronized skaters 218 (42.4%) had suffered from acute injuries during their synchronized skating career, while among 14 male skaters 6 of them (42.9%) had acute injuries while synchronized skating. The localization of acute injuries is shown in Table 1. In Tables 2–4 the spectrum of acute injuries and their localization throughout the different body regions is presented. Some of the skaters, both males and females, had suffered from more than one injury during their synchronized skating career, with median of 3–5 injuries per skater, providing a total of 398 acute injuries in females and 14 acute injuries in males. Two hundred and ninety-six (57.6%) female and 8 (57.1%) of the male skaters did not have any injury in their synchronized skating career. Reviewing the chance of overuse in-

juries within the locomotor system of these skaters we found that 69 (13.4%) female and 2 (14.3%) male skaters had suffered from low back problems and 112 (21.8%) female and 2 (14.3%) male skaters had one or more overuse syndromes during their skating career. Out of 155 overuse injuries in female skaters, 65.8% (102 overuse injuries) occurred during their figure skating career, while only 34.2% (53 overuse injuries) occurred in skaters after they joined synchronized skating teams or in those skaters who initially started their skating career as synchronized skaters. In male skaters, where we found 5 overuse injuries, 4 of them (80%) occurred in their figure skating career, while 1 (20%) occurred during his synchronized skating career. The localization and time of overuse injuries occurrence are shown in Table 5.

Synchronized skaters practice team and individual elements on and off the ice. Looking through the distribution of injuries in questionnared skaters, out of 412 injuries, 338 (82%) occurred during on-ice practice, while 74 (18%) acute injuries happened during off-ice training. Ninety-one (26.9%) acute injuries occurred while teams were practicing individual elements in their skating programs, and 247 (73.1%) injuries that happened on ice occurred while the team practiced different team elements (circle, line, wheel, intersection, block, lifts, others). Distribution of injuries through team elements is shown in Table 6.

The number of injuries increased during the last 4 skating seasons (Table 7).

The severity and type of injury determined the length of time that skaters had to interrupt their training and this time was reported between 0 days and 6 months. Injuries were treated either conservatively or with surgery. In conservative treatment different methods of physical therapy were used (RICE – rest, ice, compression, elevation, stretching, strengthening, electrostimulation of muscles, electrotherapy, laser, ultrasound, magnetotherapy, and others). Surgery was divided into two groups: major surgery and sutures for lacerations. Out of 88 laceration injuries, 65 (73.9%) required sutures for closure, while 23 (26.1%) were so superficial that they were treated with primary

**Table 2** Absolute and relative (%) numbers of acute head and trunk injuries in female (n = 514) and male (n = 14) synchronized skaters who were studied

|                    | <b>Head</b> |          | <b>Neck</b> |          | <b>Thoracic spine</b> |          | <b>Lumbar spine</b> |          | <b>Abdominal part</b> |          | <b>TOTAL<br/>no and (%)</b> |            |
|--------------------|-------------|----------|-------------|----------|-----------------------|----------|---------------------|----------|-----------------------|----------|-----------------------------|------------|
|                    | <b>F</b>    | <b>M</b> | <b>F</b>    | <b>M</b> | <b>F</b>              | <b>M</b> | <b>F</b>            | <b>M</b> | <b>F</b>              | <b>M</b> | <b>F</b>                    | <b>M</b>   |
| <b>Contusion</b>   | 23          | 1        | 3           |          | 2                     |          | 16                  |          |                       |          | 44<br>(41.1%)               | 1<br>(50%) |
| <b>Concussion</b>  | 20          | 1        |             |          |                       |          |                     |          |                       |          | 20<br>(18.7%)               | 1<br>(50%) |
| <b>Laceration</b>  | 23          |          |             |          |                       |          |                     |          | 1                     |          | 24<br>(22.4%)               | 0          |
| <b>Others</b>      | 13          |          | 2           |          |                       |          | 4                   |          |                       |          | 19<br>(17.8%)               | 0          |
| <b>Total</b>       | 79          | 2        | 5           |          | 2                     |          | 20                  |          | 1                     |          | 107                         | 2          |
| <b>No. and (%)</b> | (73.8%)     | (100%)   | (4.7%)      |          | (1.9%)                |          | (18.7%)             |          | (0.9%)                |          | (100%)                      | (100%)     |

**Table 3** Absolute and relative (%) numbers of acute upper extremity injuries in female (n = 514) and male (n = 14) synchronized skaters who were studied

|                    | Upper arm   |   | Lower arm     |   | Shoulder      |            | Elbow         |   | Wrist       |            | Finger        |            | Total of arm injuries no and (%) |             |
|--------------------|-------------|---|---------------|---|---------------|------------|---------------|---|-------------|------------|---------------|------------|----------------------------------|-------------|
|                    | F           | M | F             | M | F             | M          | F             | M | F           | M          | F             | M          | F                                | M           |
| Fracture           |             |   | 5             |   |               |            | 3             |   | 6           | 1          | 6             |            | 20<br>(15.2%)                    | 1<br>(25%)  |
| Contusion          | 8           |   | 4             |   | 8             | 1          | 4             |   | 5           |            | 6             |            | 35<br>(26.5%)                    | 1<br>(25%)  |
| Haematoma          |             |   | 3             |   | 7             |            | 3             |   | 3           |            | 2             |            | 18<br>(13.6%)                    |             |
| Laceration         |             |   | 12            |   |               |            | 5             |   | 4           |            | 11            | 2          | 32<br>(24.2%)                    | 2<br>(50%)  |
| Sprain/strain      |             |   |               |   | 4             |            | 3             |   | 11          |            | 9             |            | 27<br>(20.5%)                    |             |
| Total, No. and (%) | 8<br>(6.1%) |   | 24<br>(18.2%) |   | 19<br>(14.4%) | 1<br>(25%) | 18<br>(13.6%) |   | 29<br>(22%) | 1<br>(25%) | 34<br>(25.7%) | 2<br>(50%) | 132<br>(100%)                    | 4<br>(100%) |

**Table 4** Absolute and relative (%) numbers of acute lower extremity injuries in female (n = 514) and male (n = 14) synchronized skaters who were studied

|                    | Upper leg    |   | Lower leg     |            | Hip           |              | Knee          |              | Ankle         |            | Foot        |   | Total of leg injuries no. and (%) |              |
|--------------------|--------------|---|---------------|------------|---------------|--------------|---------------|--------------|---------------|------------|-------------|---|-----------------------------------|--------------|
|                    | F            | M | F             | M          | F             | M            | F             | M            | F             | M          | F           | M | F                                 | M            |
| Fracture           |              |   | 8             |            |               |              | 1             |              |               |            | 4           |   | 13<br>(8.2%)                      |              |
| Contusion          | 5            |   |               |            | 11            | 1            | 19            |              | 2             |            | 3           |   | 40<br>(25.2%)                     | 1<br>(12.5%) |
| Haematoma          |              |   | 2             |            | 5             |              | 12            |              |               |            |             |   | 19<br>(11.9%)                     |              |
| Laceration         | 5            |   | 11            | 2          | 4             |              | 10            | 1            |               |            |             |   | 30<br>(18.8%)                     | 3<br>(37.5%) |
| Sprain/strain      |              |   |               |            |               |              | 4             |              | 31            | 2          | 2           |   | 37<br>(23.3%)                     | 2<br>(25%)   |
| Ligamentous        |              |   |               |            |               |              | 20            | 2            |               |            |             |   | 20<br>(12.6%)                     | 2<br>(25%)   |
| Total, No. and (%) | 10<br>(6.3%) |   | 21<br>(13.2%) | 2<br>(25%) | 20<br>(12.6%) | 1<br>(12.5%) | 66<br>(41.5%) | 3<br>(37.5%) | 33<br>(20.7%) | 2<br>(25%) | 9<br>(5.7%) |   | 159<br>(100%)                     | 8<br>(100%)  |

first aid only as there was no need for any surgical intervention. The total number of fractures included 21 upper extremity fractures and 13 lower extremity fractures. There were 4 (19%) upper extremity fractures and 5 (38.5%) lower extremity fractures that required surgery.

Menarche in female skaters occurred between 10 and 15 years of age (median age 13 years old).

## Discussion

Synchronized skating is mainly a female sport and out of 23 teams who participated in this study only 6 teams reported male participants. This is the first study reporting the incidence of in-

juries in elite synchronized skaters, obtained through a questionnaire. Although the response both from males and females was 100%, the low number of males participating in this study would make any interpretation of the results pertaining to male skaters to be viewed with caution. The response of 100% is attributable to the full cooperation of team leaders, doctors, coaches, and managers. It has to be realized that a drawback of a questionnaire is that the recall of recent injuries is more accurate than from injuries suffered some years ago. Unfortunately publications on injuries in synchronized skating are lacking.

As we are not aware of published papers concerning synchronized skating injuries, we can only compare our results with data that were orally presented by Mc Carthy and Micheli at ISU Medical Congress held in Washington in March 2003 [8]. The number

**Table 5** Absolute and relative (%) numbers of overuse injuries in female (n = 514) and male (n = 14) synchronized skaters who were studied

|                            | <i>During figure skating career</i> |             | <i>During synchronized skating career</i> |             | <i>Total, no. and (%)</i> |             |
|----------------------------|-------------------------------------|-------------|-------------------------------------------|-------------|---------------------------|-------------|
|                            | <i>female</i>                       | <i>male</i> | <i>female</i>                             | <i>male</i> | <i>female</i>             | <i>male</i> |
| <i>Groin pain</i>          | 10                                  |             | 10                                        |             | 20 (12.9%)                |             |
| <i>Hamstrings syndrome</i> | 4                                   |             | 6                                         |             | 10 (6.45%)                |             |
| <i>Jumper's knee</i>       | 26                                  | 2           | 4                                         |             | 30 (19.4%)                | 2 (40%)     |
| <i>Osgood Schlatter</i>    | 15                                  | 2           | 0                                         |             | 15 (9.7%)                 | 2 (40%)     |
| <i>Shin splint</i>         | 13                                  |             | 12                                        | 1           | 25 (16.1%)                | 1 (20%)     |
| <i>Achilles tendinites</i> | 8                                   |             | 7                                         |             | 15 (9.7%)                 |             |
| <i>Plantar fascitis</i>    | 7                                   |             | 7                                         |             | 14 (9.0%)                 |             |
| <i>Ankle impingement</i>   | 3                                   |             | 7                                         |             | 10 (6.45%)                |             |
| <i>Stress fracture</i>     | 16                                  |             | 0                                         |             | 16 (10.3%)                |             |
| <i>Total, no. and (%)</i>  | 102 (65.8%)                         | 4 (80%)     | 53 (34.2%)                                | 1 (20%)     | 155 (100%)                | 5 (100%)    |

**Table 6** Distribution of acute injuries threw different region of the body and their spread through the studied synchronized skating team elements with absolute and relative (%) numbers

|                           | <i>Head</i> | <i>Trunk</i> | <i>Upper extremity</i> | <i>Lower extremity</i> | <i>Total, no. and (%)</i> |
|---------------------------|-------------|--------------|------------------------|------------------------|---------------------------|
| <i>Block</i>              | 12          | 8            | 19                     | 13                     | 52 (21%)                  |
| <i>Wheel</i>              |             |              | 6                      | 4                      | 10 (4.1%)                 |
| <i>Intersection</i>       | 10          | 4            | 21                     | 36                     | 71 (28.7%)                |
| <i>Circle</i>             | 3           |              | 7                      | 4                      | 14 (5.7%)                 |
| <i>Line</i>               | 8           |              | 7                      | 4                      | 19 (7.7%)                 |
| <i>Lift</i>               | 21          | 8            | 8                      | 8                      | 45 (18.2%)                |
| <i>Others</i>             | 12          | 2            | 12                     | 10                     | 36 (14.6%)                |
| <i>Total, no. and (%)</i> | 66 (26.7%)  | 22 (8.9%)    | 80 (32.4%)             | 79 (32%)               | 247 (100%)                |

of injuries as found in the present study are in line with those as presented by Mc Carthy and Micheli [8] who reported an injury rate of 29.1% at the ISU World Challenge Cup in Boston 1996 and 62.9% injuries at the United States Figure Skating Association (USFSA) Synchronized Skating Nationals in 1998. They reported a low prevalence of head injuries (4.6%), in general with no specific details of the severity or type of injury, in the ISU World Challenge Cup in 1996 and 1.2% in USFSA Nationals in 1998, while we found 19.8% head injuries in female and 14.3% in male skaters, who were classified into 4 groups – concussions, contusions, lacerations, and other as shown on Table 2. Out of 81 head injuries only 12 (14.8%) of them occurred more than 4 years ago, while 69 (85.2%) occurred in the last 4 skating seasons. This suggests that due to increasing demands put on synchronized skaters in the last 4 skating seasons, the incidence of head injuries is increasing. However, it cannot be ruled out that the recall of recent injuries is more accurate than those suffered some years ago. Head injuries occurred mainly during the practice of on-ice team elements, especially in lifts, blocks, and intersections.

Smith and Ludington [11] reported in elite pairs and ice dance skaters 11 serious injuries that were caused by lifts. In previous research [5] in elite junior figure skaters, head injury occurred in 8 (13.1%) females and 5 (8.2%) male pair skaters, and in 1

**Table 7** The incidence of acute injuries with absolute and relative (%) numbers due to time of its occurrence in studied female (n = 514) and male (n = 14) synchronized skaters

|                                     | <i>Less than 4 years ago</i> |                         | <i>More than 4 years ago</i> |             | <i>Total</i>  |             |
|-------------------------------------|------------------------------|-------------------------|------------------------------|-------------|---------------|-------------|
|                                     | <i>no. and (%) female</i>    | <i>no. and (%) male</i> | <i>no. and (%) female</i>    | <i>male</i> | <i>female</i> | <i>male</i> |
| <i>Head, no. and (%)</i>            | 67 (16.8%)                   | 2 (14.3%)               | 12 (3%)                      | 0           | 79 (19.8%)    | 2 (14.3%)   |
| <i>Trunk, no. and (%)</i>           | 17 (4.3%)                    | 0                       | 11 (2.8%)                    | 0           | 28 (7.1%)     | 0           |
| <i>Upper extremity, no. and (%)</i> | 83 (20.85%)                  | 3 (21.4%)               | 49 (12.3%)                   | 1 (7.2%)    | 132 (33.15%)  | 4 (28.6%)   |
| <i>Lower extremity, no. and (%)</i> | 87 (21.85%)                  | 5 (35.7%)               | 72 (18.1%)                   | 3 (21.4%)   | 159 (39.95%)  | 8 (57.1%)   |
| <i>Total, no. and (%)</i>           | 254 (63.8%)                  | 10 (71.4%)              | 144 (36.3%)                  | 4 (28.6%)   | 398 (100%)    | 14 (100%)   |

(1.5%) male ice dancer. Brown and McKeag [3] reported that lift-related injuries in pair skaters mostly included contusions, lacerations, fractures, haematomas, and muscle strains. In this research we found that among synchronized skaters reported head injuries included not only contusions, haematomas, and lacerations, but also the incidence of concussions.

Among trunk injuries, the majority were located over the lumbar spine (Table 2). Mc Carthy and Micheli [8] reported 20.2% of upper extremity, 58.4% of lower extremity, and 14.5% of trunk injuries in USFSA Synchronized Skating Nationals, while we found 33% of upper, 40.5% of lower extremity, and 6.8% of trunk injuries with major prevalence in the last 4 skating seasons, as shown on Table 7. Most of the trunk, upper, and lower extremity injuries that occurred on ice, occurred while the skaters were practicing intersection and block team elements, which require a great deal of coordination, unison, and precision to execute successfully. Reviewing the overall lower extremity injuries in synchronized skaters, the majority of them are knee injuries (Table 4), mostly ligament/meniscus injury, contusions, haematomas, and lacerations, while in a previous study [5] we found that among pairs and ice dancers knee injuries are a minority compared to the other acute injuries. In figure skaters, among acute injuries, the prevalence of ankle sprains is greater than 50%, as reported by Authrosen [1], Bloch [2], Danowski [4], and Dubravcic-Simunjak et al. [5], while in the present study we found that in synchronized skaters the incidence of ankle sprains/strains was only 19.5% in female and 25% in male skaters. During their training figure skaters practice double, triple, and quadruple jumps on and off the ice, and due to failures in clean landings ankle sprains occur quite often in that skating discipline. In synchronized skating, jumps are limited to single rotations in which ankle sprains usually do not occur. However synchronized skaters are subject to ankle sprains when they are practicing some of the team elements, mostly off the ice, but these elements are at a much lower risk of ankle sprain than the risk associated with multirotational jumps. Hence, the rate of ankle sprains is much lower in synchronized skating than in figure skating.

Of all upper extremity injuries recorded in synchronized skaters, the most commonly found were finger injuries, followed by wrist injuries and then by lower arm injuries being the third most common upper extremity injury reported (Table 2). In a previous study [5] in junior figure skaters, out of 258 pair skaters and ice dancers, only 17 (6.6%) reported upper extremity injuries, while none was found among 211 single men and women figure skaters. The injuries recorded in this study of synchronized skating are easily comprehended given the large number of skaters simultaneously on the ice performing different elements and holds presenting demanding elements synchronously. Most of the upper and lower extremity injuries occurred in intersections in which one part of the team moves through or along the remaining part of the team. The second most common circumstance where injuries occur are when the skaters are moving as units along the ice in blocks in varying shapes of three or more lines.

In synchronized skating overuse injuries are not as common as acute ones. Although 155 overuse injuries in female and 5 in

male skaters were reported, most of them occurred before those skaters joined synchronized skating teams (Table 5). In this respect synchronized skaters are similar to ice dancers and pair skaters in which acute injuries are more common than overuse ones [5,10,11]. In single figure skaters (men and women) most of their training time is spent on training 6 different triple jumps and one or two of those as quadruples. Due to uncontrolled repetitions and microtrauma of lower extremities overuse injuries are seen in those skaters more often than the acute ones [5].

Pairs and ice dancers spend most of the training time on practicing different pair elements – lifts, throws, pair spins, steps, and holds. To gain extra credit and points, those elements have to be original and performed in unison, both in pairs and in ice dancers. Synchronized skaters work in their practice on different team elements (blocks, circles, intersections, wheels, lifts...) which require good posture, skating skills, coordination, and timing. In spite of the many repetitions they do not lead to great impact or microtrauma on the lower extremities. So, the most likely explanation for the occurrence of the two different types of injury is the difference in training routine between those skating disciplines.

Our results show that the number of low back problems are similar to those found in other figure skating disciplines [2,4,5,9]. Omeý and Micheli [9] found that besides overuse and acute injuries, low back pain was the most frequent complaint seen in singles and pair figure skaters.

Looking through the injuries that occurred in surveyed skaters, we noticed that the best-placed teams (medal winners) had lower rate of injuries than teams placed from 4th to 23rd place.

Limitations of this study are its retrospective design and lack of data from other studies reporting synchronized skating injuries, however, this is the first one presenting injury data in this skating discipline.

In conclusion, we would like to stress that the current available data suggests that the number of acute injuries has increased in the past 4 skating seasons, although it cannot be ruled out that the numbers have been biased by a better recall of recent injuries. Although it is tempting to suggest that the increase coincides with increased demands for more technically difficult elements performed by the synchronized skaters, this conclusion should be treated with caution. Nevertheless, we strongly recommend that the medical community work closely with the coaches, skaters, and technical committees in future development of synchronized skating.

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